

PIUM EMPIRICAL GALAXY ROTATION CURVE RE-ANALYSIS

Structural Pattern of the Outer Rotation-Curve Slope Distribution

Author: Szabolcs Krehlik
ORCID: 0009-0003-8623-7876
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STRUCTURAL PATTERN OF THE OUTER SLOPE DISTRIBUTION

The distribution of outer rotation-curve slopes in the accepted SPARC subsample exhibits a clear internal structure rather than a purely random scatter. In the present recalculation, the accepted analysis sample contains 79 galaxies selected under the criteria ($Q \leq 2$), ($R > 2.5 R_{\text{disk}}$), and ($N_{\text{outer}} \geq 8$). The analysed quantity is the logarithmic outer slope

$$s \equiv d \log_{10}(V_{\text{obs}}) / d \log_{10}(R)$$

measured by ordinary least squares regression in log–log space across the defined outer region.

The resulting slope distribution is characterized by a mean of 0.044600, a median of -0.000632 , and a standard deviation of 0.176476, with values ranging from -0.217107 to 0.613538. Operationally, galaxies are classified as declining for $s < -0.05$, flat for $-0.05 \leq s \leq 0.05$, and rising for $s > 0.05$. Under this definition the sample contains 28 declining galaxies, 18 flat galaxies, and 33 rising galaxies.

At first glance these category counts might suggest three distinct dynamical populations. However, the statistical structure of the distribution indicates a different interpretation. The most striking feature of the sample is that its central tendency lies extremely close to zero slope: the median value is effectively null within the precision of the measurement. This implies that the characteristic outer dynamical state of the accepted SPARC galaxies is not systematically rising or systematically declining, but instead clustered around a nearly flat rotation-curve configuration.

This observation is important because the flat outer rotation curve has historically been treated largely as a qualitative statement about galaxy dynamics. In the present recalculation it emerges quantitatively as the central state of the distribution. At the same time, the distribution exhibits a substantial width, with a standard deviation of approximately 0.176. Individual galaxies therefore depart noticeably from this central regime. The empirical picture is not one of perfect universality, but rather of a preferred dynamical state surrounded by structured deviations.

A second notable feature is the asymmetry between the mean and the median. The mean slope is positive, whereas the median remains essentially zero. This indicates that the distribution is not symmetric about its central value. Instead, the positive side of the distribution contains a longer or more influential tail produced by galaxies with strongly rising outer curves. The sample is therefore skewed toward positive slopes. This asymmetry carries astrophysical significance because it implies that deviations from flatness are not randomly distributed in sign. A subset of galaxies retains measurable rising behaviour even in the outer regions defined by the present selection criteria.

The operational category counts reinforce this interpretation. Although the flat interval defines the central classification band, it is not the numerically dominant category in the accepted sample. The sample instead contains substantial declining and rising populations, with the rising class slightly exceeding the declining class. The practical implication is that the widely cited statement that outer rotation curves are “flat” should be interpreted as describing the central tendency of the distribution rather than a universal rule applying to all galaxies.

The third and most informative structural feature appears in the relationship between outer slope and asymptotic rotation velocity. The recalculated sample reveals a statistically significant negative correlation between s and V_{flat} , with Pearson coefficient $r = -0.625803$ and Spearman rank coefficient $\rho = -0.660607$. Both correlations correspond to very small probability values, indicating that the observed trend is unlikely to arise from random fluctuations in the sample.

This result demonstrates that the outer slope is not an isolated descriptive quantity but a scale-dependent dynamical parameter. Galaxies with larger asymptotic rotation velocities tend to exhibit flatter or declining outer slopes, whereas galaxies with smaller V_{flat} are more likely to show rising outer behaviour. The outer rotation profile therefore depends systematically on the dynamical scale of the galaxy.

Several consequences follow from this correlation. First, the slope distribution cannot be interpreted simply as a mixture of unrelated rotation-curve morphologies. Instead, the data suggest a continuous dynamical progression across galaxy scale. Second, the common declining/flat/rising trichotomy is best understood as an operational classification applied to an underlying continuum. The categories remain useful for descriptive statistics, but they do not imply the existence of three sharply separated physical populations.

The sensitivity analysis further supports this interpretation. When the outer-region definition is varied, the resulting statistics change in a coherent and systematic manner. For $R > 2.0 R_{\text{disk}}$, the accepted sample contains 87 galaxies and the mean slope equals 0.073798. For the fiducial definition $R > 2.5 R_{\text{disk}}$, the sample contains 79 galaxies and the mean slope becomes 0.044600. When the outer boundary is pushed farther outward to $R > 3.0 R_{\text{disk}}$, the sample decreases to 65 galaxies and the mean slope declines to 0.007398. A final method based on the last 40 percent of radial points produces a sample of 66 galaxies with a mean slope of 0.035615.

These systematic shifts provide important physical insight. As progressively more external regions of the rotation curve are isolated, the average slope of the sample moves toward zero. In other words, the outer rotation field tends toward a nearly flat configuration when sufficiently large galactocentric radii are considered. This does not imply that every galaxy becomes perfectly flat, but it does indicate that the ensemble behaviour of the outer population approaches a near-zero logarithmic slope at large radius. The coherent behaviour of the statistics under different outer-region definitions also indicates that the recovered structure is not a numerical artifact of the calculation pipeline.

The radial acceleration relation recalculated for the accepted sample provides an additional consistency check. Using 2338 data points from the selected galaxies, the RAR regression yields a slope of 0.659997, a correlation coefficient of 0.953082, and an RMS scatter of 0.169158 dex. These values reproduce the well-known dynamical regularity of the SPARC dataset while confirming that the recalculation pipeline preserves the fundamental scaling relations of galaxy dynamics.

Taken together, the results reveal a coherent empirical structure in the outer slope distribution. The accepted SPARC galaxies cluster around a nearly flat central regime, but the departures from this regime are not random. Instead they form a continuous scale-dependent sequence. Low- V_{flat} galaxies preferentially occupy the positive-slope side of the distribution, intermediate systems concentrate near the flat state, and higher- V_{flat} galaxies increasingly populate the near-flat or mildly declining region.

The outer slope can therefore be interpreted as a structural dynamical observable that reflects the position of a galaxy along a broader sequence of dynamical scale. Under this interpretation, the outer slope is not merely a descriptive property of rotation curves but a measurable parameter encoding information about the dynamical state of galactic systems. This empirical framework provides a suitable baseline for future comparisons with theoretical models of galaxy dynamics, including both conventional gravitational descriptions and alternative field-based approaches.